

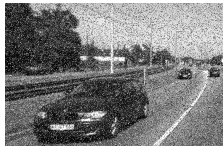
Abstract

In this lab, I implemented a version of the Adaptive Median Filter (AMF) using an Integral Histogram. I evaluated the efficacy of this approach by comparing it to both the Gaussian filter and the normal variant of the Median filter. The Mean Square Error (MSE) of the resulting images compared to the original image was used as an empirical measure of the noise restoration.

Introduction

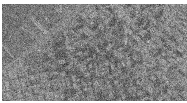
I first developed a function that takes in two images (an original and a restored version), then calculates the MSE of these images. I applied this to a test image with two different levels of salt and pepper noise restored through various library methods to gain a baseline for my AMF. I then developed a function to generate an Integral Histogram of an image and built my AMF off of that. I measured its effectiveness on the test case from part 1 as well as a more complicated image.

Mean Square Error

MSE	Unprocessed	Gaussian Filter Sigma = 2	Gaussian Filter Sigma = 7	Median Filter (7x7)	Median Filter (19x19)
Test1Noise1					
Test1Noise1	21.781	92.391	97.806	36.992	52.019
Test1Noise2					
Test1Noise2	65.085	106.617	103.643	42.803	53.083

- While all of the filters removed some amount of salt and pepper noise, the results were generally unacceptable.
- The Gaussian filter is generally inappropriate to use for images affected by salt and pepper noise. With a low sigma, it only manages to spread the noisy pixels, causing the grainy effect seen above. A higher sigma can remove the noise, but only as a byproduct of removing all detail from the image. The noisy pixels are not discernable in the high sigma images above, though the result is unacceptable due to the loss of detail in the image.
- The Median filter was far more successful in restoring these images, particularly with a smaller kernel. It has a similar consequence to the Gaussian filter of smoothing the image, but since the process is more dependent on intensity values, more detail is retained.
 - The (7x7) filter is best for the lower noise of Test1Noise1. It removes all noisy pixels and retains a good amount of detail, although the image is notably smeared.
 - For Test1Noise2, there is such a high level of noise that the (7x7) filter fails to remove it all. This indicates that at certain points, the majority of pixels in the kernel are noise. The (19x19) does better, but completely decimates the image's detail.
- Notably, all of the filtered images have a *higher* MSE than the unprocessed image for Test1Noise1. While this indicates that none of the filters are particularly effective, it also demonstrates the limitations of MSE as a measurement. Direct mathematical comparison of pixel intensity values are not how humans engage with images, and more information can be gleaned from the de-noised image passed through the (7x7) median filter (with the background trees illustrating this best).

The Adaptive Median Filter

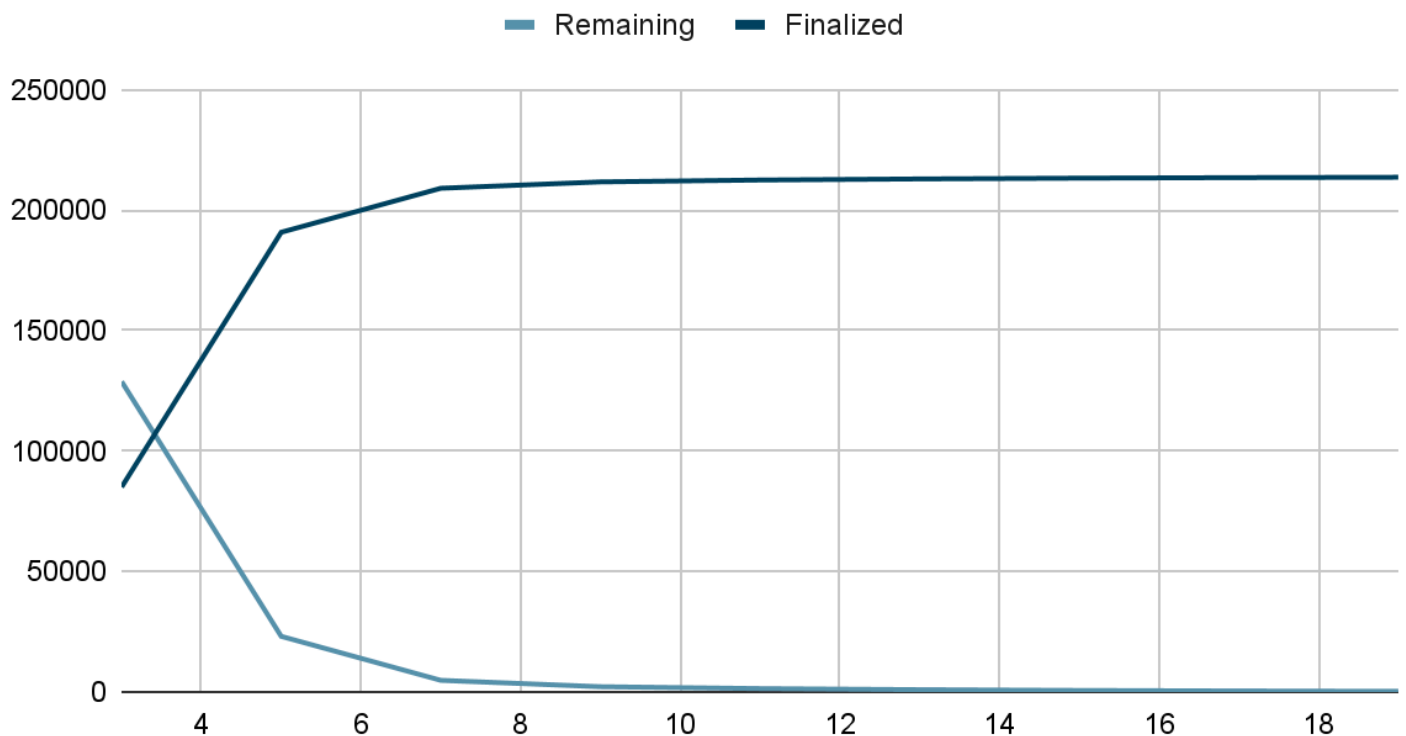
		MSE		Processing Time (s)	
	Noisy	Adaptive Median Smax = (7x7)	Adaptive Median Smax = (19x19)	Adaptive Median Smax = (7x7)	Adaptive Median Smax = (19x19)
Test1 Noise 2					
Test1 Noise 2	65.085	35.640	35.654	36.343	36.827
Test2 Noise 2					
Test2 Noise 2	76.837	62.046	61.836	2:23.461	2:27.139

- My implementation of the Adaptive Median Filter is based on the concept of the Integral Histogram, where a histogram is generated for every pixel containing the intensity values of the pixels above and to the left. This dramatically decreased the processing time for each window size, as a histogram could be generated for any region with a simple equation. This improvement is illustrated by the negligible time difference in execution for different maximum window sizes. The increase from 7 to 19 added another .5 seconds for Test 1 and 3.5 seconds for Test 2.
- Coupled with the dramatic decrease in processing time for applying the AMF is the large time expenditure required to develop the Integral Histogram. I utilized the scanning wavefront method described in the original paper, where a histogram is first created for the pixel in the origin, and each new pixel's value is added to this as the scan moves outwards. This was far more efficient than creating a histogram wholesale for every pixel. Still, ~80% of the program's processing time is dedicated to generating this Integral Histogram. A naive approach likely would have been faster, as these are not complicated enough tests to necessitate the Integral Histogram.
- My implementation was generally successful, with much lower MSEs than any of the filters used in part 1. The filter does not remove any detail, except in the rare situation where a normal filter has the max/min intensity value of a noise pixel.

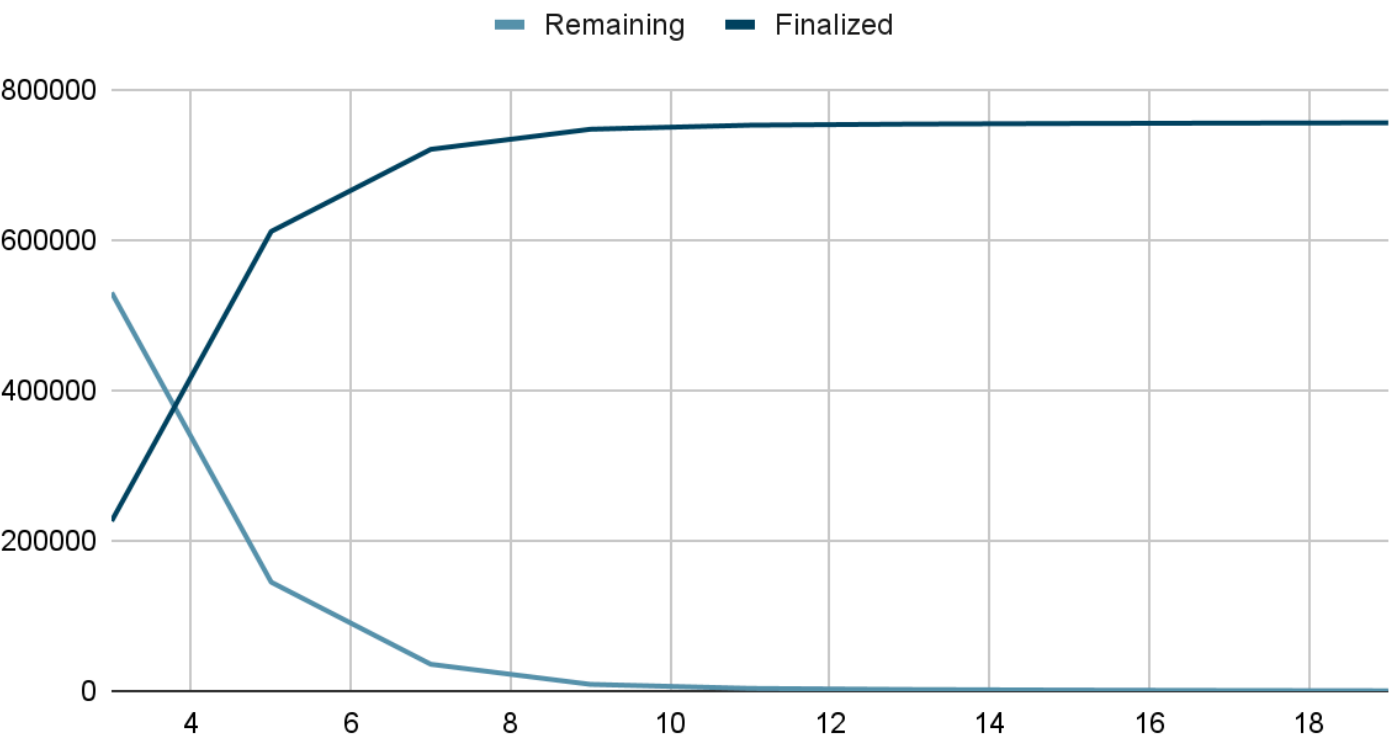
- Test 2 was especially successful, as the image goes from indistinguishable from static to recognizable as London. The main issue, as in Test 1, is that the filter fails to remove all noise pixels. This is due to the high concentration of noise. For certain pixels, their neighborhood will be composed primarily of noise regardless of the window size. Once the window reaches its maximum size, they will default to the value of the majority of noise pixels.
 - This is another downside of my Integral Histogram approach. Normally, I would be able to use already reconstructed pixels in median calculations for other pixels. Sadly, there is no way to update an Integral Histogram that does not require an absurd amount of computation. Any new pixel would have to be updated in the histograms for every pixel below and to the right of it. This means that certain noisy pixels simply cannot be removed.
- The Integral Histogram was the incorrect approach for this purpose. I believed it would save processing time overall, but did not consider how its inflexibility would limit the AMF's effectiveness.

The below graphs plot the number of remaining and finalized pixels for each window size for tests 1 and 2. The same general trend is seen in both, where the majority of pixels are reconstructed after the first pass through.

Pixel Status at Each Window Size for Test 1



Pixel Status at Each Window Size for Test 2





AMF 7



AMF 19



AMF 7



AMF 19